Prashant Govindarajan

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Education

- **Mila and Polytechnique Montréal**, Montréal, Canada — *Aug'22-Present*
  PhD in Computer Engineering; Advisor: Sarath Chandar
- **Indian Institute of Technology Madras**, Chennai, India — *Jul'17-Jul'22*
  BS in Biological Sciences and MTech in Data Science; Minor: Computational Biology — *CGPA: 9.31/10*

Research Areas and Interests

Reinforcement Learning, Large Language Models, AI-based Material and Drug Discovery, Geometric Deep Learning, and Computational Biology

Relevant Coursework & Skills

Courses: Geometry and Generative Models, Reinforcement Learning, Representation Learning, Robot Learning, Parameter and State Estimation, Parallel Scientific Computing, Algorithmic Approaches to Computational Biology, Pattern Recognition and Machine Learning
Skills: Python (PyTorch, Tensorflow, JAX), R, MATLAB, C/C++ (OpenMP, MPI, OpenACC), Matter Modeling (DFT)

Professional Experience

- **Software Engineering PhD Intern, Google**
  Location: Bengaluru, India — *May'26-Present*
  - Designing agentic systems for automated semantic and data discovery in Google Cloud's global infrastructure team
  - **Skills**: Agentic system design frameworks (e.g., Google ADK), testing and evaluating agents

Research Experience

- **Reinforcement learning for LLMs**
  Mentors: Sarath Chandar, Saurav Jha — *Sep'25-Present*
  - Improving exploration with intrinsic rewards in RL-based fine-tuning of LLMs
- **AI-automated CAD object generation**
  Mentors: Sarath Chandar, Jay Pathak, Quentin Fournier — *Jun'24-Jan'26*
  - Collaboration with Ansys: LLMs for automated generation of 3D CAD objects
  - Published CADmium, an open-source dataset and LLM fine-tuning approach at TMLR [Paper, Code, Website]
- **Reinforcement learning for material design**
  Mentors: Sarath Chandar, Mathieu Reymond, Santiago Miret, Mariano Phielipp — *Aug'22-Present*
  - Project with Intel on offline and online RL approaches for generating new crystal structures
  - Integrated first-principles DFT with conservative Q-learning – accepted at MoML 2023, AI4Mat at NeurIPS 2023, and Digital Discovery [Paper, Code]
  - Released CrystalGym, first online RL environment for material discovery – Spotlight at AI4Mat-ICLR 2025 [Paper, Code]
  - Currently working on LLM fine-tuning with RL for crystal generation
- **Master's Thesis: Graph generative models for binding site-specific molecule generation**
  Guides: Balaraman Ravindran, Karthik Raman, IIT Madras [Thesis, Poster] — *Aug'21-Jun'22*
  - Designed graph VAE models for drug molecule generation; explored RNN/LSTM for node and edge generation with mitigated order dependence
- **Analysis of drug response and gene expression data of AML cells**
  Guide: Brian Wilhelm, Université de Montréal — *Jun-Sep'21*
  - Computational methods to identify drug-gene correlations and molecules inducing leukemic cell maturation
- **Deep generative models for single-cell gene expression analysis**
  Guide: Hongyu Zhao, Yale University (Virtual) — *May-Jul'20*
  - Evaluated VAEs and other unsupervised deep learning for single-cell gene expression data analysis
- **RNA-seq data analysis of human oral squamous cell carcinoma**
  Guide: Debnath Pal, Indian Institute of Science Bangalore — *May-Jul'19*
  - Identified somatic mutations in RNA-seq data of human oral squamous cell carcinoma samples

Projects

- **Effects of visual representation for navigation control tasks**
  Robot Learning, Université de Montréal *Jan-Apr'23*
  - Studied effects of contrastive learning and VAE-based pretraining for RL-based visual navigation
- **Incorporating geometry into score-based model for crystal structure design**
  Geometry and Generative Models, McGill University *Sep-Dec'22*
  - Incorporated crystal symmetry as inductive bias into generative models for crystal structure design
- **Generating drug-like molecules from gene expression signatures using transformer**
  Algorithmic Approaches to Computational Biology, IIT Madras [Poster, Video, Report] *Sep-Dec'20*
  - Attention-based transformer for *de novo* drug-like molecule generation inducing desired transcriptomic profiles. Poster at MLCSB COSI, ISMB 2022
- **Parallel analyses of canonic polyadic tensor decomposition algorithm**
  Parallel Scientific Computing, IIT Madras [Report, Code] *Feb-Jun'21*
  - CPU- and GPU-level parallelization of tensor decomposition using OpenMP and OpenACC
- **Deep generative approach to model single-cell data of human embryoid bodies**
  Computational Systems Biology, IIT Madras *Jan-Jul'20*
  - Used deep generative VAE model (scVI) to identify biologically relevant cell types of single-cell human embryoid bodies

Awards

- **FRQNT Doctoral Scholarship 2026** – Merit scholarship awarded by Fonds de recherche du Québec
- **PBEEE 2025** – Merit scholarship for international students awarded by Fonds de recherche du Québec
- **Khorana Program for Scholars 2020** – Awarded by Dept. of Biotechnology, Govt. of India (research internship fellowship)
- **INSPIRE Scholar** – Awarded by Department of Science and Technology, Government of India

Publications (Google Scholar)

- **Govindarajan, Prashant**, Davide Baldelli, Jay Pathak, Quentin Fournier, and Sarath Chandar. **"CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design."** arXiv:2507.09792 (2025). ***Accepted at TMLR***
- **Govindarajan, Prashant**, Mathieu Reymond, Antoine Clavaud, Mariano Phielipp, Santiago Miret, and Sarath Chandar. **"CrystalGym: A New Benchmark for Materials Discovery Using Reinforcement Learning."** arXiv:2509.23156 (2025).
- **Govindarajan, Prashant**, Santiago Miret, Jarrid Rector-Brooks, Mariano Phielipp, Janarthanan Rajendran, and Sarath Chandar. **"Learning Conditional Policies for Crystal Design Using Offline Reinforcement Learning."** Digital Discovery (2024).
- **Govindarajan, Prashant**, Kishalay Das, and Sarath Chandar. "CrysTune: Crystal Generation via Fine-Tuning of Large Language Models on Wyckoff Representations." ICML 2026 AI for Science Workshop.
- Woo, Kowen, **Prashant Govindarajan**, and Sarath Chandar. **"Benchmarking Machine Learning Potentials for Crystal Structure Relaxation."** NeurIPS 2025 AI for Science Workshop.
- **Govindarajan, Prashant**, Mathieu Reymond, Santiago Miret, Mariano Phielipp, and Sarath Chandar. **"Crystal Design Amidst Noisy DFT Signals: A Reinforcement Learning Approach."** AI for Accelerated Materials Design-NeurIPS 2024.
- **Govindarajan, Prashant**, Santiago Miret, Jarrid Rector-Brooks, Mariano Phielipp, Janarthanan Rajendran, and Sarath Chandar. **"Behavioral Cloning for Crystal Design."** ML4Materials workshop, *ICLR 2023*.

  Accepted Posters: AI4Science (ICML 2026 & NeurIPS 2025), AI4Mat (ICLR 2025, Vienna 2024, NeurIPS 2023 & 2024), ML4Materials (ICLR 2023), MoML 2023, ISMB 2022

Activities

Teaching

- Machine Learning, Polytechnique Montréal, Fall 2025 | Reinforcement Learning, Polytechnique Montréal, Fall 2023
- Reinforcement Learning, IIT Madras, Spring 2022 | DSA for Biology, IIT Madras, Fall 2021

Activities

- Information Security Committee at Mila
- Organizer of AI for Materials reading group at Mila
- Instructor for Chandar Lab's High School Internship Program
- Volunteer for Graduate Application Assistance for Underrepresented Students in AI
- Organizer of MoML 2023 and MoML 2024 at Mila | Social events organizer at Chandar Lab

Reviewing

- TMLR (2025), AI4Science (NeurIPS 2025, ICML 2026), AI4Mat workshop (Vienna 2024, NeurIPS 2023 & 2024), MoML (2023-25), Deployable AI (AAAI 2023)

Competitions

- First prize in start-up pitch at Sciencepreneurship, EPFL | Winning team, MIT COVID-19 Challenge | Finalist, NYAS Tracking Coronavirus

Others

- Summer School Acceptances: Oxford ML (2024, declined), Sciencepreneurship (2024), Amii AI Week (2022, online)
- Ultimate Frisbee (NSO, IIT Madras) | Coordinator, Sponsorship team, Shaastra 2019, IIT Madras | Coordinator, Analytics Club, CFI, IIT Madras